From Norm-Five Quaternions to Primitive Cycles: Exact Nonbacktracking Dynamics of the LPS $X^{5,q}$ Family

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Abstract

For every prime $q > 5$ with $q \equiv 1 \pmod 4$, six Hamilton quaternions of norm five define a six-regular Lubotzky–Phillips–Sarnak Cayley graph. We close its arithmetic and dynamical specialization in one normalization. The Legendre symbol $(5/q)$ selects a connected $\mathrm{PSL}_2(\mathbb{F}_q)$ or $\mathrm{PGL}_2(\mathbb{F}_q)$ chamber and exactly controls bipartiteness. The oriented-edge Hashimoto operator has a complete Bass characteristic factorization, all-iterate trace formula, Möbius primitive-cycle inversion, and finite-graph Ihara product.

Keywords: arithmetic Cayley graphs; LPS graphs; Hashimoto operator; nonbacktracking dynamics; primitive cycles; finite-graph Ihara product

中文摘要

我们从范数五的六个整数四元数出发，把它们约化为有限域上的
投影矩阵，并冻结右乘的Cayley规范。若素数 $q > 5$ 且
$q \equiv 1 \pmod 4$，则平方类 $(5/q)$ 把同一构造分入
$\mathrm{PSL}_2(\mathbb{F}_q)$ 与 $\mathrm{PGL}_2(\mathbb{F}_q)$ 两个算术室。前者非二分，
后者由行列式平方类给出等大的二分部；群阶、六度性以及更换
$\sqrt{-1}$ 所引起的共轭等价均被明确确定。 在此图上提升到有向边后，禁止立即逆行的转移给出一个内生动力学。
我们专门化Bass–Hashimoto行列式，固定所有次幂迹的
闭步约定，并用Möbius反演恢复每个长度的本原有向闭轨；
相应的有限图Ihara乘积因而逐项确定。
关键词： 算术图动力学；LPS图；非回溯转移；有向闭轨；
Hashimoto算子；有限图Ihara乘积

1 Arithmetic quaternion chambers

Let

$$\mathcal{S}_5 = \{(1, \pm 2, 0, 0), (1, 0, \pm 2, 0), (1, 0, 0, \pm 2)\}.$$

Choose $\iota \in \mathbb{F}_q$ with $\iota^2 = -1$ and put

$$M_\iota(a) = \begin{pmatrix} a_0 + \iota a_1 & a_2 + \iota a_3 \\ -a_2 + \iota a_3 & a_0 - \iota a_1 \end{pmatrix}. \tag{1}$$

We use its projective class. Write $\chi_q = (5/q)$ and

$$G_q = \begin{cases} \mathrm{PSL}_2(\mathbb{F}_q), & \chi_q = 1, \\ \mathrm{PGL}_2(\mathbb{F}_q), & \chi_q = -1. \end{cases}$$

Lemma 1 (Norm, inverses, and gauge)**.** *The six classes $[M_\iota(a)]$ are distinct and inverse closed. Replacing ι by $-\iota$ conjugates the generating set and gives an isomorphic labeled Cayley dynamics.*

Proof. Expansion gives

$$\det M_\iota(a) = a_0^2 + a_1^2 + a_2^2 + a_3^2 = 5.$$

Quaternion conjugation changes the sign of the unique nonzero tail coordinate and satisfies $M(a)M(\bar{a}) = 5I$. The paired projective classes are inverses. For $q > 5$, equality of two classes, or equality with the identity class, would force one of $2, 4, 5$ to vanish modulo q. Thus the six classes are distinct and nonidentity. For $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, direct multiplication gives $M_{-\iota}(a) = J M_\iota(a) J^{-1}$. ∎

Theorem 1 (LPS chamber closure)**.** *The Cayley graph $X^{5,q} = \mathrm{Cay}(G_q, \mathcal{S}_5)$ is connected, simple, six-regular, and Ramanujan. More precisely,*

$$|G_q| = \begin{cases} q(q^2-1)/2, & \chi_q = 1, \\ q(q^2-1), & \chi_q = -1. \end{cases}$$

The first chamber is nonbipartite. The second is bipartite, with the square and nonsquare projective determinant classes as its two equal parts.

Proof. Connectedness, the group identification, and the Ramanujan bound are the LPS theorem specialized to quaternion prime five [1]. Lemma 1 gives simplicity and degree six, while the displayed orders are the standard projective-group orders. Projective determinant square class is well defined because scalar rescaling changes determinant by a square. If $\chi_q = -1$, every edge flips this character; connectedness and one generator give a bijection between the two parts. If $\chi_q = 1$, a connected bipartite Cayley graph would define a nontrivial word-parity map $\mathrm{PSL}_2(\mathbb{F}_q) \to \{\pm 1\}$. Its kernel would have index two, contradicting the simplicity of $\mathrm{PSL}_2(\mathbb{F}_q)$ for $q \geq 13$. □

2 Exact primitive nonbacktracking dynamics

Let $n_q = |G_q|$ and $m_q = 3n_q$. On the $2m_q = 6n_q$ oriented edges, define $H_q(e, e') = 1$ when the terminus of e is the origin of e' and $e' \neq \bar{e}$. Let A_q be adjacency on vertices.

Theorem 2 (Complete determinant and orbit ledger)**.** *One has the polynomial identities*

$$\det(tI - H_q) = (t^2-1)^{2n_q} \det(t^2 I - tA_q + 5I), \tag{2}$$

$$\det(I - uH_q) = (1-u^2)^{2n_q} \det(I - uA_q + 5u^2 I). \tag{3}$$

If $N_q(r) = \mathrm{Tr}(H_q^r)$ and $\Pi_q(r)$ counts prime oriented cycles of length r, modulo cyclic shift but not reversal, then

$$\Pi_q(r) = \frac{1}{r} \sum_{d|r} \mu(d) N_q(r/d). \tag{4}$$

Moreover, in $\mathbb{Q}[[u]]$ and as a rational function,

$$\prod_{[C]} (1 - u^{\ell(C)})^{-1} = \exp\left(\sum_{r \geq 1} \frac{N_q(r)}{r} u^r \right) = \det(I - uH_q)^{-1}. \tag{5}$$

Proof. The Bass–Hashimoto formula for a graph with degree matrix D is

$$\det(I - uH) = (1-u^2)^{m-n} \det(I - uA + u^2(D - I)).$$

Here $D = 6I$ and $m - n = 2n$, which proves (3); substitution $u = t^{-1}$ proves (2) [2, 3, 4]. By construction, a diagonal entry of H_q^r selects a closed oriented-edge walk with no inverse pair, including across the terminal seam. Thus its trace counts cyclically nonbacktracking walks. Every such walk is a unique repeat of a prime oriented cycle, so $N_q(r) = \sum_{d|r} d\Pi_q(d)$. Möbius inversion proves (4); grouping all repeats, or taking the formal logarithm of the finite determinant, proves (5). □

References

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